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Agentic AI

04 February 2026

Hotel Support Bot Proposal

Problem Statement

Hotels usually receive a high volume of repetitive customer inquiries pertaining to check-in and check-out times, parking, amenities, pets, etc. In most situations, these questions are handled by front desk staff or call centers which leads to long wait times and inconsistent results for the customer. A hotel support bot would solve this problem by providing accurate and consistent answers to customer questions in a reasonable amount of time while also reducing the workload of the staff.

The target users for this for this support bot are primarily hotel guests and potential customers looking for information about hotel services and policies. It can also be used by the hotel staff to refer to policies whenever necessary.

An agentic solution is appropriate here because it will allow the agents to really understand customer requests and provide the correct information. Agents will be able to process user questions based on what is asked and determine the best course of action. A predefined external workflow would only be able to answer simple and straightforward questions whereas an agentic system can take a customer question and understand what they mean and respond appropriately.

Use Cases and Scenarios

Scenario 1 : A guest wants to know policies before booking a room

Example: “What time is check-in and what time is check-out?”

Tasks performed by the Agents:

- Classify the intent of the query. In this case its related to room policy.
- Retrieve check-in and check-out information from the hotel room data
- Generate a professional response for the customer
- Review the response for correctness
- Convert to audio response for user

Scenario 2 : A guest wants to know about amenities

Example: “Do you have free breakfast and parking?”

Tasks performed by the Agents:

- The query would be identified as an amenity related question
- Retrieve breakfast and parking information from hotel data
- Generate a professional response for the customer
- Check answer for correctness
- Convert to audio response for user

Initial System Design

Agent List:

- Identifier/Classification Agent

This agent identifies the category of the user request. For example, it'll take a question and decide if it's related to room policy, amenities, accommodations, pets, parking, etc.

- Rule Agent

This agent will store all of the rules, policies, and guidelines of the hotel.

- Retrieval Agent

This agent will retrieve relevant information from the hotel policies and guidelines based on the category selected by the previous agent.

- Response Agent

This agent will be responsible for converting the information retrieved by the knowledge agent into a response for the user.

- Critic Agent

This agent will review the generated response for correctness and completeness. If any errors are detected, edits will be made here.

- Voice Agent

This agent will convert the information into an audio format for the end user.

Expected Tools:

- Local File System

This will store hotel policies and guidelines for the system to reference when answering guest questions or requests.

- Large Language Model

An LLM is necessary to be able to understand natural language and to be able to generate responses. A locally hosted model would be beneficial because it would reduce privacy concerns and overall costs.

Feasibility and Risks

Potential Challenges:

- Misclassification

If a user query is broad or general, the system may not be able to identify the information needed for the request. This can be mitigated by giving the model limited categories to select from and having a option for human escalation if the model is unable to help with the request.

- Hallucination

It is possible the system generates solutions that are not within the company policies or guidelines. This can be helped by requiring the model to retrieve information only from predefined hotel rulebooks.

- Tool Failure

File read errors could occur when retrieving hotel information.

Constraints:

- Privacy

There are always risks and privacy concerns if using customer data.

- Time

The development of this AI bot must be completed within the duration of this course.

Timeline:

- Week 1: Proposal
- Week 2: Milestone 1 – Front-end
- Week 3: Milestone 2 – Multi-Agent Workflow
- Week 4: Milestone 3 – Evaluation Plan and Pilot Results

- Week 5: Peer Rating / Feedback
- Week 6: Final Presentation
- Week 7: Final Implementation and Report